On Design and Implementation of Gamed-Based Learning Modules for Graph Theories

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Abstract

In this paper, a set of modules as the standards for design and develop the games for learning the graph theory are describes. The game-based learning is a new approach for learning. It is feasible for developing the game for learning the graph theories. We introduce a set of modules as the fundamental elements for designing and developing the learning game for graph theories.

1 Introduction

This paper describes our attempts of designing and implement the gamed-based learning modules for teaching graph theories. The initialization of this study is from the third author’s popular book[6]. For the educational purposes, more supplemental teaching materials for intriguing the young readers in graph theories is an important issues. In additions, the game-based learning becomes a popular topics for online teaching and eLearning research. We plan to initiate the study of developing the game-based learning materials for graph theories.

Generally, the graphics theories and discrete structures are the fundamentals of computer science. However, except for college courses the materials for teaching the graphics theories are very scarce. There are many important areas can apply the graph theories. For example, one important graph theory, the Hamiltonian path problem are applied to many different applications [6][13].

To describe this problem for the general readers, a popular science book with many vivid descriptions for learning the Hamiltonian path problem is published. In the book, the Hamiltonian path problem was described as the routes and mazes in the zoo. With careful design, this book is polished with many interesting examples for general readers even the juvenile.

The young readers are all digital natives. According to the study[7], the characteristics of these digital natives are the followings:

1. The digital natives frequently use the mobile devices such as IPad or iPhone;
2. The digital native are familiar with the computer and network technologies;
3. The digital natives frequently use web applications and mobile phone based technologies.

To make this group of readers to understand and interest this problem, the digital version of the original book should be designed and created for these digital natives. In addition, the juvenile paly the video game are heavily. Of course, there is another important conjecture: the young digital native might understand what the popular items are for other young people. Currently, the video games and apps for the handheld devices are used for these young people. The development of video games with the content of graph theories would be popular. Previously, we had guided students to develop their own version of the Hamiltonian path apps.

In the previously study, we guided the college students who are computer-related major to develop the Hamiltonian path app. The app would be more illustrative for other young people. The notion of this idea fit to the important HCI design concept, the participatory design process. After finishing our first prototype implementation, we had accumulate some important findings and information for game designs. We found the followings:

1. The participatory process can help these involving students learn the graph theories; students becomes more interesting in the theories and applications of the graph theories.
2. Students would like to add more information that they are familiar into the design of the app. One group of student put some famous Grimms’ Fairy Tales as the stories to lead to the game challenge of

the Hamiltonian path apps. The other team put the lyrics and melody of a popular Taiwanese band, sodagreen into their design as the successors' rewards.

3. The biggest challenges of the design process is how to design the scenarios and the interface to put all the contents together to become an app.
4. From the design process, the participants learns the integration of the app design and the discrete graph theories. This encourage us to develop more projects based on this preliminary project.

Based on these findings, we concludes some future research trends:

1. How to integrated the graph theories into the learning process? There should be some guidelines or proposed structures.
2. Could we use the theories and practices of the game-based learning to develop our implementations? The possible solutions should be studies and developed.
3. For these participants, how to develop the scenarios and solutions for implement the graph theories app?
4. The methods for evaluation of the game should be developed.

In this paper, from the above conclusions, we propose a framework for designing and implement the game-based learning modules for the graph theories. Based on the development of game-based learning and related study, we design a set of modules that can be used for designing the games to learn the graph theories. This design is used for designing the game-based learning material for graph theory and for helping other college students who plan to study the graph theories and would like to implement a game about the graph theories.

The remaining of this paper is organized as the follows. Section 2 gives a brief description of the methodology and theorems of the game-based learning used in our proposed modules. Section 3 is the description of how we use the game-based learning theory to design games for graph theories. Section 4 describes the proposed modules to design the game. Section 5 describe the game is implemented following the modules. Conclusions and future study are in the final section.

2 Theories for Gamed-Based Learning

What are the purposes for using the game to learn these graph theories? Based on the book of James Paul Gee[4], the games can bring several key issues into the learning process. These issue are designed and put in the form of texts, sounds, music, movement and bodily sensations. These are be coined as "semiotic domain." The learners are taught to understand the meanings in the domains. The components in the game should be design as modules in the "semiotic domain" and can be grouped into different playgrounds for the game-based learning. In the design of the game, the HCI for game design are used. The game design is a very new HCI research topic. The players of the game become the main focus in the study of the game design. Therefore many strategies to find the profiles and perspective of the players have been developed[10][13]. The research of the game design focus on four aspects: learning, storytelling, game play and user experience. The game design related study showed that how to design criteria and framework for evaluating the previous four aspects are the mainstream research topics in game design. In this paper, we will put these considerations into the design of our process.

Another methodology we use in the design is the knowledge space theory. This is the theory used for eLearning design [12] or for diagnosis for the states in learning mathematics. Usually, the knowledge is ordered into different sets. Among these sets, some are labeled as the prior knowledge sets and the others are labeled as the posterior knowledge sets. The prior knowledge sets are prerequisite before the posterior knowledge sets. Hence, the sequence of these knowledge sets will be organized as the learning schedule. The learning path can be design by using the knowledge space theory.

3 Modules for Game Design

How to use the game for learning the graph theories. The learning foci of the design is to develop the method for acquiring the knowledge/concept of the graph theories. Here, we describe the considerations and the design following a specific consideration for this purpose.

The modules in the design are regards the kernel of the whole game designs. Based on the modules, the game can be expanded and implemented. The game evaluation can based on the defined modules to get the game ideas. Unlike the flowchart element, the modules includes several different elements across game design, software design and learning model.

Based on the theories and the learning model, the modules in the game design for learning the graph theories are (1) the player, (2) learning core activities, (3) learning challenges and (4) rewards. The module, as the kernel, are design as the first consideration of the game design. When a game design process starts, the designer will collect information and consider the modules. In the very early stage, the designer should consider the module the player.

The participants in the game, the payers, are the people who are the targets of the game. The players can enjoy the games and learning the knowledge from the game. The goal of the game is to set up a plan to help the player to learn certain knowledge. Based on different learning model[11], different strategic methods are used for develop the game. AT the beginning of the learning game design, the description of the players should be provided. The designers will exam these descriptions and find the best strategy to design the game.

The learning core activities are regards as the processes in the game for deploying the learning materials. The learning core activities are transform into the roles of plays in the game. Therefore, the learning core activities are regards as what the designers or instructor want to teach the players and what are the steps for the instructors to conduct the teaching. The core activities can be grouped as the complex role play or can be used as a simple play instruction. The core activities should strongly linked to the knowledge that plan to teach the players.

The learning challenges are the challenges and difficulties added to the game. The challenges can be treated as the barriers that slows the players' path to finish the game. The challenges can be another place for deploying the learning knowledge. Some advance knowledge can be put in learning challenges, such that the players will take time to solve the problems and get the information. Sometime, another set of information can be put in the learning challenges.

The rewards are the animations, pictures, songs and music given to the players when the players finish a stage of play or the whole play. The rewards can be used as the reminiscence of the learning knowledge. The players will find intriguing and learning again from the rewards.

The design of the game will use these modules to design the whole framework of the game. The modules can used as the following: (1) the preliminary descriptions of the game and to collect the required learning for design the games; (2) guidelines for the design and implement: the modules can be treated as the guidelines to conduct the detailed design and coding. (3) Assessment: the modules can be used as the references to develop the assessment methods.

4 Implementation

Based on the proposed framework, we had instructed a group of seniors in the Computer Science and Communication Engineering Department to develop an games for learning graph theories. The original ideas are from the book, Mathematics in Zoo[6]. The **players** of the game are the 5^{th}-grade-to-9^{th}-grade students who interests in playing video games. The knowledge set for the game design includes the graph theories and the Taiwanese ecological areas which are exhibits in the National Natural Science Museum. The **learning core activity** of the game is to finish drawing the graphs in the graph theory. Each different graph has its rules for drawing. The **rewards** of the game are the ecological information and its description. The **learning challenges** are the difficulties and levels of these drawing graph.

In this game system, the Taiwanese ecology is put in a series of three different games. In each game, an ecological area is linked with one specific graph theory to group and design the game. At the beginning, the main menu page (Figure 1) appears and prompts to asking the player to choose a game. Once the payer picks up a game, he will enter the game world and playground which are designed as an Taiwanese ecological area. The scenarios of games are described as the follows:

Game1: **The An-ma-shan cloud forest(Figure 2):** the environment regards to the warm and rainy forest in mid-elvated Taiwan mountains. The plants and animals live in this area are these familiar to local students. The graph game is Euler graph(Figure 3,Figure4, Figure5). Players in this game should finish the assigned Euler graph to get the information of this ecology area. Once the player finishes one round, he can unlock the next round (Figure 6). After the player finish all rounds of the game, he will get a reward, a picture and description of one animal or plant in this ecological area (Figure 7).

Game2: **The Nan-hu valley,** a unique frigid valley located in southern Taiwan. The special plants and animals always intrigue many visitors. The graph game here is the famous Hamiltonian graph. Players should use the interactive tool to complete trace all the edges and vertices in the Hamiltonian path. Six different Hamiltonian graphs are used in this game with different difficulties. Player who only finish each graph can get the interesting information of the Naa-hu valley. Each animal for one graph.

Game3: **The Lian-hua-chih broad-leaved forest**: the environment that represent the ecological uniqueness of Taiwan. Plants and animal are unique in Taiwan around the world. In this game the bipartite graph is used for design the graph game. The more challenges comes with different bipartite graphs. To help the player understand the bipartite graph, the part of sub graph in different sets is colored with different color.

5 Conclusion

In this paper, we propose a set of modules for design and develop the game for learning the graph theories. The development is still under way to finish and the pilot test showing the design modules are efficient for helping to develop the game for learning the graph theories.

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Figure 1 The main menu

Figure 2 The An-ma-shan Forest

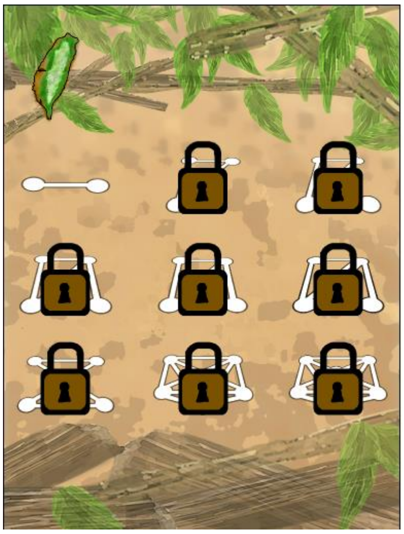

Figure 3 Graphs play menu

Figure 4 Information of Euler path

Figure 5 Euler path playground

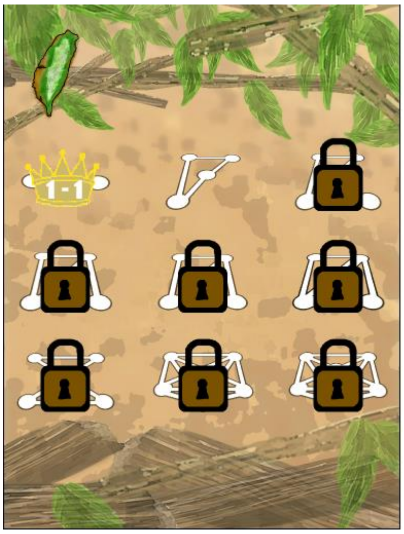

Figure 6 Complete one graph

Figure 7 Reward Picture

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